

Department of Statistics and Data Science, The Chinese University of Hong Kong

Lecture 5

Fundamental Finance for Risk Managers and Actuaries

Programme: *Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students*

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Opening Problem

An insurance company holds a portfolio containing Treasury bills, long-term bonds, listed shares, and several derivative contracts used for hedging. At the same time, it has promised to pay future claims whose timing and size are uncertain. A new analyst is asked to prepare a briefing for the chief risk officer. The briefing must explain four things clearly. First, what kinds of markets make these transactions possible? Second, why do bonds, equities, and derivatives play different roles on the balance sheet? Third, how should future obligations be translated into present-day values? Fourth, why do formulas such as e^{rt} and e^{-rt} keep appearing in modern finance and actuarial work? The analyst quickly realises that the questions are inseparable. A good answer demands a single coherent framework linking markets, instruments, risk, valuation, and continuous-time thinking.

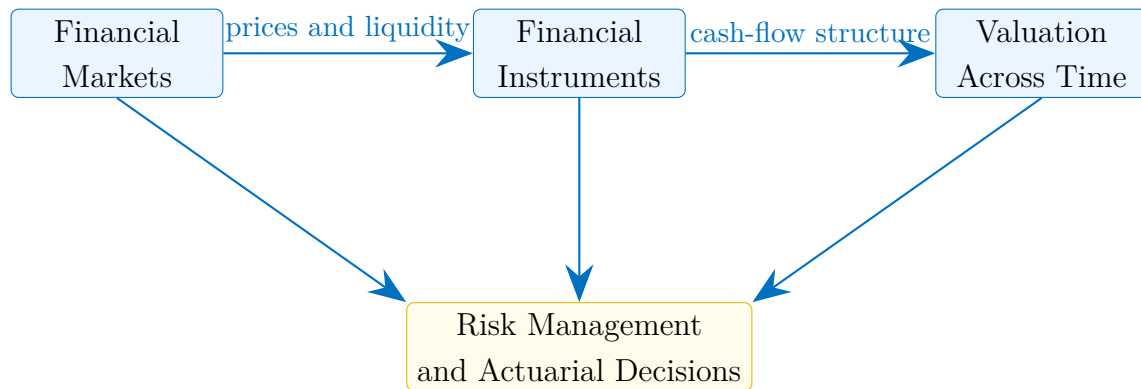


Figure 1: The organising logic of Lecture 5: the chapter moves from financial markets to instruments to valuation, and then draws those ideas together for actuarial and risk management decisions.

Interpretive Note

A useful way to read this chapter is to follow one repeated question: *what cash flows does this instrument generate, and how should those cash flows be valued today?* Markets matter because they produce prices and liquidity. Instruments matter because they determine who receives which cash flows and with what degree of certainty. Valuation matters because all later judgments about investment choice, reserve adequacy, and solvency depend on translating those cash flows onto a common present-value scale. This repeated question is what holds the chapter together.

1 A. Financial Markets as the Environment of Risk Management

1.1 A1. Why markets matter

Definition and Interpretation

A financial market is an organised mechanism through which financial claims are issued, exchanged, priced, and revalued. For a risk manager or actuary, a market is not merely a venue for speculative trading. It is a system that supports price discovery, liquidity, and risk transfer.

Financial markets matter because valuation and risk management do not occur in a vacuum. If an insurer invests collected premiums, it must place those funds in actual instruments traded in actual markets. If a pension fund discounts liabilities, the chosen rates must be interpreted in relation to market yields. If a company hedges interest-rate,

exchange-rate, or commodity risk, it must rely on markets in which derivative positions can be entered, monitored, and adjusted. The market is therefore not a decorative background to finance. It is part of the financial problem itself.

A professional view of markets begins with three broad functions. The first function is **price discovery**. Markets aggregate judgments and information, and transaction prices become reference points for valuation. The second is **liquidity**. A liquid market allows assets to be sold or purchased quickly without a severe price concession. The third is **risk transfer**. Markets make it possible for one party to reduce an exposure by shifting part of that exposure to another party willing to bear it.

Table 1: Core market functions and their practical significance for risk managers and actuaries

Market function	What the function means	Why the function matters in practice
Price discovery	The market generates observable prices for securities through trading and information aggregation.	Risk managers need reference prices and yields in order to value portfolios, estimate capital strain, and assess the financial effect of changing conditions.
Liquidity	Investors can buy or sell assets quickly without extreme price disruption.	Insurers and other institutions may need cash quickly to pay claims, collateral, or obligations. An illiquid portfolio may look solvent on paper and still be dangerous in practice.
Risk transfer	Exposures can be passed or shared among many participants.	Hedging, portfolio rebalancing, and diversification depend on the ability to shift risk through markets rather than bearing every exposure directly.

Worked Example

Why liquidity can matter as much as theoretical value? (Core) Imagine that an insurer holds a large amount of wealth in an investment that is difficult to sell quickly. In calm times, the theoretical valuation may appear satisfactory. However, if a sudden catastrophe produces a wave of claims, the company may need cash immediately rather than a theoretical long-term value. If that asset cannot be sold without a large discount, the firm faces a serious practical liquidity problem even though the balance sheet looked acceptable before the event. The lesson is that value and liquidity are related but distinct ideas. Sound risk management must always consider both.

1.2 A2. Primary markets and secondary markets

Key Definition / Result

Primary markets are markets in which new securities are created and sold for the first time. **Secondary markets** are markets in which existing securities are bought and sold among investors after issuance. Issuance determines original contract terms, while secondary trading determines liquidity and ongoing market value.

When a government issues a new Treasury bill, or a company issues new shares or bonds, the transaction belongs to the primary market. At that stage, maturity, face value, coupon structure, and contractual rights are established. Once issued, the same security may later trade many times in the secondary market. Those later trades do not send new capital to the original issuer, but they are essential for liquidity and for the market valuation used by financial institutions.

Common Mistake

Confusing issuance with later trading A common misunderstanding is to think that every transaction in a security is part of the issuer's fundraising process. That is not correct. Fundraising occurs in the primary market, while later investor-to-investor transactions occur in the secondary market. Risk managers care strongly about secondary markets because portfolio valuation and liquidity depend heavily on them.

1.3 A3. Money markets and capital markets

Money markets specialise in short-term debt instruments, usually with maturities of one year or less. Capital markets deal with longer-term debt and equity claims. This distinction is useful because liability timing should influence asset choice. Short-term reserve needs and near-term liabilities are often linked to short-term assets because those assets are generally more liquid and less exposed to long-horizon uncertainty. Long-tail liabilities may require an asset strategy that looks further ahead and balances liquidity, income, and return potential more carefully.

Interpretive Note

A useful professional habit is to ask not simply, “What asset offers the highest return?” but rather, “What asset structure best supports the timing, liquidity, and risk characteristics of the liability I am trying to manage?” This habit turns finance from a search for isolated yield into a disciplined exercise in matching purpose to structure.

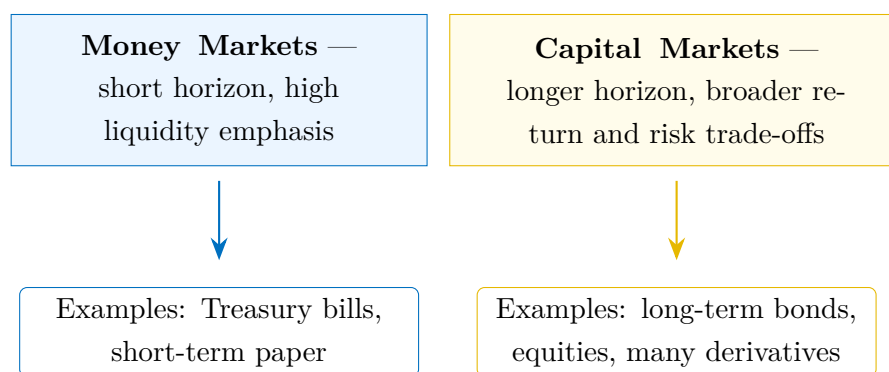


Figure 2: A simple horizon-based view of the market landscape. The distinction is useful because timing is central to valuation and liability support.

2 B. Financial Instruments as Tools for Allocation, Valuation, and Control

2.1 B1. Debt instruments and contractual cash flows

Definition and Interpretation

A debt instrument is a financial claim under which the issuer promises specified payments to the investor under agreed terms. Common examples include Treasury bills, notes, bonds, and corporate debt. Their defining feature is that the timing and amount of at least some payments are contractually structured.

Debt instruments matter especially to insurers, pension funds, and many institutional investors because predictable payment patterns are useful when liabilities are also time-structured. This does not mean that debt is risk-free. Bonds still face default risk, interest-rate risk, inflation risk, and reinvestment risk. The point is that the structure of the claim is more explicit than in equity. That structure allows analysts to value the security by discounting promised cash flows and to discuss how well the asset supports future obligations.

Table 2: Selected instrument types and their professional interpretation

Instrument	Core cash-flow idea	Typical risk emphasis	Common professional use
Treasury bill	Purchased at a discount and repaid at face value at maturity.	Low credit risk, short-horizon reinvestment and rate considerations.	Liquidity reserve, short-term asset allocation, benchmark short rate reference.
Coupon bond	Periodic coupon payments plus principal repayment at maturity.	Interest-rate risk, credit risk, inflation risk, duration exposure.	Liability support, income generation, asset-liability matching.
Equity share	Residual claim on the business, with uncertain dividends and capital gains.	Volatility, systemic risk, business risk, valuation uncertainty.	Long-term growth, surplus participation, return enhancement.
Derivative contract	Value linked to an underlying asset, price, or rate.	Model risk, basis risk, leverage effects, counterparty or market risk.	Hedging, exposure adjustment, tactical risk transfer.

2.2 B2. Equity instruments and the logic of residual claims

Equity represents ownership rather than lending. A shareholder is not promised fixed repayment in the way that a bondholder is. Instead, the shareholder participates in the residual financial outcome of the enterprise. If the company prospers, the shareholder may benefit through capital gains and dividends. If the company performs poorly, the shareholder may experience substantial losses. This makes equity a natural example of speculative risk.

Opening Problem

Company XYZ collapses and enters liquidation. After the sale of its remaining assets, only HK\$10 million is left. Bondholders are owed HK\$8 million. Shareholders originally invested HK\$20 million of equity capital. Who is likely to be paid, and why does anyone buy equity if bondholders have priority? This question is not merely about legal order. It is also a question about risk, reward, and the structure of financial claims.

Worked Example

Interpreting the capital stack (Insight) In a simplified capital-structure view, bondholders are creditors. They have a prior contractual claim on the existing assets of the company. Shareholders are residual claimants. This means that shareholders receive what remains after creditors have been paid. In the scenario above, if HK\$10 million remains and bondholders are owed HK\$8 million, bondholders would first be paid up to the amount contractually due, leaving HK\$2 million as the residual amount available to equity holders. Equity investors accept this junior position because they also participate in upside if the company's value rises substantially. Equity is therefore a high-uncertainty claim with potentially high rewards, whereas debt is a more structured claim with greater priority.

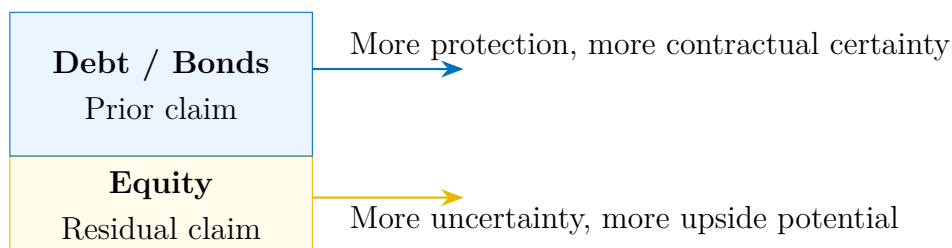


Figure 3: A simplified capital stack. Debt stands higher in priority, while equity absorbs greater uncertainty in return for potential upside.

2.3 B3. Derivatives and deliberate risk transformation

Derivative instruments derive their value from reference variables such as underlying asset prices, interest rates, exchange rates, or indexes. Their professional importance lies in the fact that they allow exposures to be modified without always changing the underlying holdings directly. A portfolio exposed to falling bond prices may use derivatives to hedge rate sensitivity. A company exposed to commodity prices may use futures to reduce uncertainty. A business with foreign exchange exposure may use forwards or options to manage that risk.

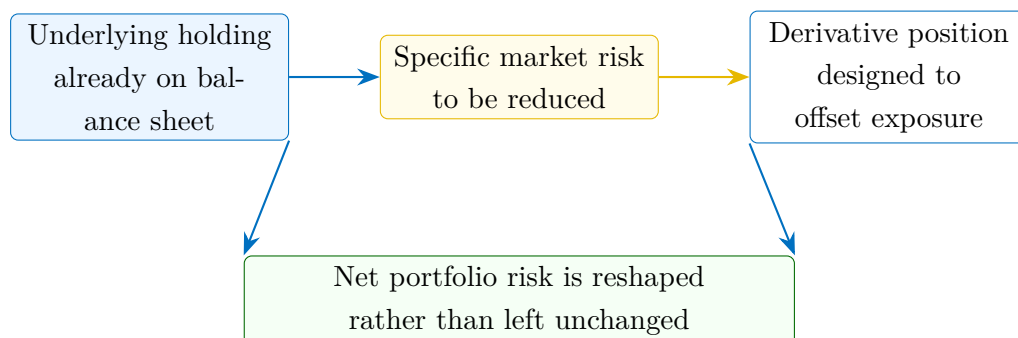


Figure 4: A conceptual diagram of hedging. Derivatives matter because they allow existing exposures to be adjusted rather than merely accepted.

Investigation / Activity

Hedging versus speculation. Imagine that a portfolio manager holds a large amount of long-term Treasury bonds. If interest rates rise sharply, bond prices will tend to fall. The manager decides to sell Treasury futures as a hedge. Explain in complete sentences how this creates a protection layer rather than a second unrelated gamble. Then explain why hedging is often described as socially productive, whereas pure speculation may not be.

AI Exploration Prompt

Use an AI assistant to test your explanation. A suitable prompt would be: “*Explain to a secondary student why a risk manager who already owns long-term bonds might sell Treasury futures. Compare the financial outcome when interest rates rise with and without the hedge, and explain why this is different from speculation.*”

Common Mistake

Assuming that all derivative use is speculative. A derivative can certainly be used for speculation, but speculation is not its defining professional purpose. In risk management, a derivative is often introduced specifically to reduce an existing exposure. The instrument is therefore best understood in relation to the portfolio it modifies.

3 C. Time Value of Money as the Engine of Valuation

3.1 C1. Why valuation across time is indispensable

Whenever finance asks whether an asset is worth buying, whether a reserve is sufficient, or whether a future payment is large or small relative to current resources, it is asking a time-value question. A dollar today and a dollar years from now are not economically identical because current money can be invested and future money must be discounted before it can be compared with present resources. Time value of money is therefore not a side topic. It is the central mechanism of valuation.

3.2 C2. Discrete compounding

Key Definition / Result

If principal P grows at nominal annual rate r , compounded m times per year for t years, the accumulated value is

$$A = P \left(1 + \frac{r}{m} \right)^{mt}.$$

The formula captures repeated multiplication by a growth factor over many equal time periods.

The formula should be understood, not merely memorised. The term r/m is the periodic rate, and the exponent mt counts the total number of compounding periods. Growth is multiplicative. Interest earned in early periods becomes part of the base on which later interest is earned.

Worked Example

Discrete compounding step by step (Core) A fund of HK\$120,000 is invested at a quoted annual rate of 6%, compounded monthly, for three years. We identify $P = 120,000$, $r = 0.06$, $m = 12$, and $t = 3$. Substituting into the formula gives

$$A = 120,000 \left(1 + \frac{0.06}{12}\right)^{36}.$$

The monthly rate is 0.005, so the factor inside the bracket is 1.005. When this factor is raised to the power 36, the result is approximately 1.19668. Multiplying by the principal gives about HK\$143,601.60. The increase over time is larger than under simple interest because the interest itself earns interest when compounding is used.

Table 3: One-year growth factors under the same quoted annual rate of 8%

Convention	Formula	Growth factor
Annual compounding	1.08	1.080000
Quarterly compounding	$(1 + 0.08/4)^4$	1.082432
Monthly compounding	$(1 + 0.08/12)^{12}$	1.083000 (approximately)
Daily compounding	$(1 + 0.08/365)^{365}$	1.083278 (approximately)
Continuous compounding	$e^{0.08}$	1.083287 (approximately)

3.3 C3. Present value as the reverse of accumulation

If accumulation moves money forward in time, discounting moves money backward to the present. In discrete form,

$$PV = \frac{F}{(1 + i)^t}$$

translates a future amount F into its present equivalent at effective annual rate i . This formula lies behind bond pricing, reserve valuation, project appraisal, and many other applications.

Worked Example

Present value of a future payment (Core) Suppose that a liability of HK\$500,000 is due in five years and the relevant effective annual discount rate is 4%. The present value is

$$PV = \frac{500,000}{(1.04)^5}.$$

Since $(1.04)^5 \approx 1.21665$, the present value is approximately HK\$410,964. This means that under the stated rate, an amount of about HK\$410,964 today is financially equivalent to a payment of HK\$500,000 five years from now.

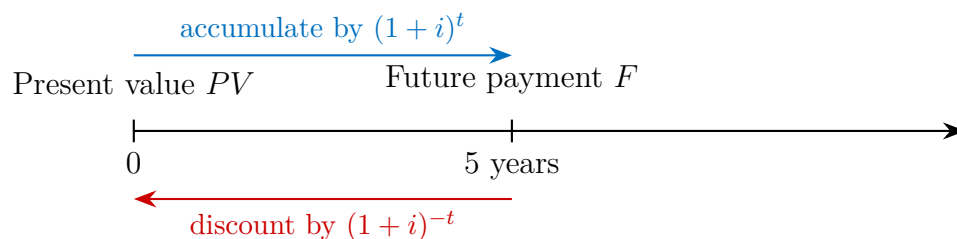


Figure 5: A valuation timeline. Accumulation moves present funds forward, while discounting translates future cash flows back to the present.

3.4 C4. From time-value formulas to the valuation of bonds and stocks

The previous formulas become genuinely useful only when they are attached to real instruments. A bond is valued by discounting the contractually promised coupons and principal. A stock is valued by discounting the cash that shareholders expect to receive, usually through dividends or through future distributions that depend on earnings and growth. The crucial point is that the *same time-value language* lies behind both instruments, even though the certainty of the cash flows is very different. Bonds are usually easier to value because their promised payments are more explicit. Stocks are harder because future dividends and growth are uncertain.

Key Definition / Result

Bond valuation principle. The value of a default-free bond is the present value of its promised coupons and principal repayment. For a bond with annual coupon C , face value F , maturity n , and yield y ,

$$P_{\text{bond}} = \frac{C}{(1+y)} + \frac{C}{(1+y)^2} + \cdots + \frac{C}{(1+y)^n} + \frac{F}{(1+y)^n}.$$

Stock valuation principle. A stock is the present value of the future cash flows expected by shareholders. Under the constant-growth dividend model, if next year's dividend is D_1 , the required return is k , and the expected constant growth rate is g with $k > g$, then

$$P_0 = \frac{D_1}{k - g}.$$

This stock formula is a simplified benchmark, not a universal law. It is useful because it makes the valuation logic transparent: price rises when expected cash flow rises, and price falls when the required return rises.

Table 4: Why bond valuation and stock valuation feel different even though both use present value

Instrument	What is discounted	Main valuation driver
Coupon bond	Contractual coupon payments and principal repayment.	Market yield, maturity, coupon structure, and credit quality.
Equity share	Expected shareholder cash flows, often proxied by future dividends or sustainable earnings power.	Earnings expectations, growth expectations, and the return required by investors.

Worked Example

Valuing a bond by discounting its promised cash flows (Core) Consider a three-year bond with face value HK\$1,000, annual coupon rate 5%, and market yield 4%. The annual coupon is HK\$50. The bond price is therefore

$$P = \frac{50}{1.04} + \frac{50}{(1.04)^2} + \frac{1,050}{(1.04)^3}.$$

Calculating each present value gives approximately 48.08, 46.23, and 933.57. Adding them gives

$$P \approx 1,027.88.$$

The price is above face value because the coupon rate 5% is higher than the market yield 4%. In plain language, the bond promises cash flows that are slightly more attractive than what the market currently requires, so investors are willing to pay a premium.

Worked Example

Valuing a stock with a simple constant-growth dividend model (Insight) Suppose a company is expected to pay a dividend of HK\$2.40 next year. Investors require a return of 8%, and dividends are expected to grow at 3% per year indefinitely. Under the constant-growth model,

$$P_0 = \frac{2.40}{0.08 - 0.03} = \frac{2.40}{0.05} = 48.$$

So the model values the stock at about HK\$48 per share. The interpretation is important. The stock price becomes higher if expected dividends rise or if growth becomes stronger. The stock price becomes lower if investors demand a higher return because the business is judged to be more risky. This example is intentionally simple, but it makes a central professional idea visible: stock valuation is driven by uncertain future cash flows rather than by a fixed contractual repayment schedule.

Common Mistake

Treating stock valuation and bond valuation as identical tasks Both bonds and stocks are valued using present value ideas, but they should not be described as equally certain. A bond usually gives more explicit promised payments, so the main challenge is choosing an appropriate discount rate and interpreting market yield. A stock often requires explicit assumptions about future dividends, growth, or earnings. The mathematics of discounting is shared, but the informational problem is not.

Reasoning Gateway

This subsection is the missing bridge between the earlier instrument discussion and the later exponential discussion. Once students see that a bond price and a stock price are both present values of future cash flows, the later appearance of e^{rt} , e^{-rt} , and log returns becomes easier to motivate. Continuous-time formulas do not replace valuation logic; they refine it.

4 D. From Repeated Compounding to Euler's Number e

4.1 D1. The compounding paradox

Opening Problem

Suppose that a bank offers an annual rate of 8% on a HK\$100,000 emergency reserve. One version compounds once a year, one compounds every month, and one compounds more and more frequently. A natural question then arises. If we keep increasing the compounding frequency without bound, does the final amount become unbounded as well? Or does the process move toward a stable benchmark? This is the compounding paradox that leads directly to Euler's number.

To understand why Euler's number appears in finance, begin with the benchmark expression

$$\left(1 + \frac{1}{m}\right)^m.$$

This expression arises if we set principal equal to one, time equal to one year, and rate equal to 100%. When we evaluate it for larger and larger values of m , the values rise, but they rise more slowly each time. The sequence appears to approach a stable limit.

Key Definition / Result

The benchmark approached by the repeated-compounding sequence

$$\left(1 + \frac{1}{m}\right)^m$$

is Euler's number:

$$e \approx 2.718281828 \dots$$

In finance, the important point is that e arises naturally when compounding becomes more and more frequent.

Reasoning Gateway

Two effects happen simultaneously as the compounding frequency m increases. First, the number of compounding periods increases, which pushes the amount upward. Second, the interest rate applied in each period becomes smaller because the annual rate is divided across more periods. The key insight is that these two forces do not produce infinite growth over a fixed horizon. Instead, they interact in a balanced way and lead to a stable continuous-time benchmark. That benchmark is expressed through e .

Table 5: Observed values of $(1 + 1/m)^m$ for increasing m

m	1	2	4	12	52	365	1000
$(1 + 1/m)^m$	2.0000	2.2500	2.4414	2.6130	2.6926	2.7146	2.7169

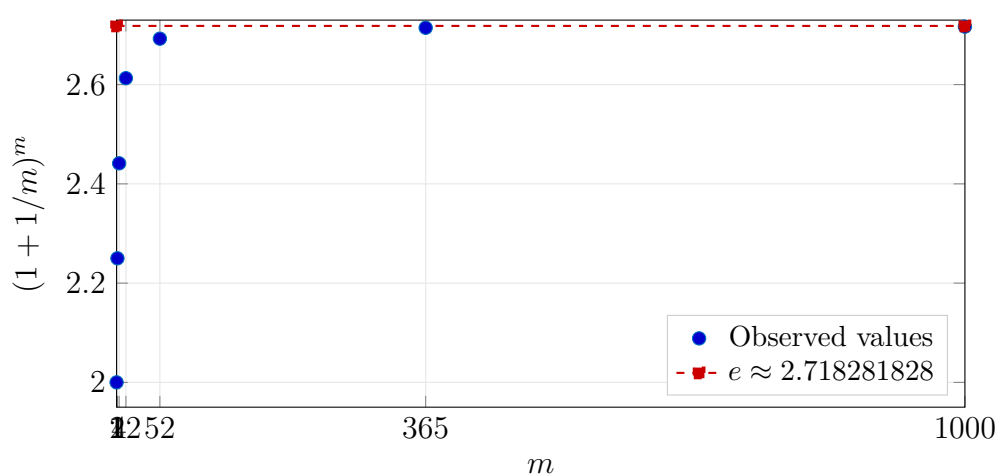


Figure 6: A visual representation of the repeated-compounding experiment. The observed values move steadily toward the benchmark e .

Common Mistake

Misreading frequent compounding as unlimited growth It is natural to think that if compounding happens more frequently, the final amount should explode without limit. For a fixed annual rate over a fixed horizon, that is incorrect. The amount increases, but it approaches a stable upper benchmark rather than growing without bound. Euler's number captures that benchmark.

5 E. Continuous Compounding, Continuous Discounting, and the Natural Logarithm

5.1 E1. Continuous accumulation

Key Definition / Result

If a quantity with present value P grows at continuous annual rate r for t years, its future value is

$$A(t) = Pe^{rt}.$$

This is the continuous-compounding counterpart of the ordinary accumulation formula.

Continuous compounding is not always meant as a literal description of how money is credited at every instant in a bank account. It is better understood as a natural limiting model and a mathematically elegant benchmark. It simplifies analysis, links directly to logarithms, and provides a compact language for smooth growth.

Worked Example

Working example: the correct continuous model (Core) An insurer holds HK\$50,000 growing at a **continuous** annual rate of 6%. After four years, the future value is

$$A = 50,000e^{0.06 \times 4} = 50,000e^{0.24}.$$

Using technology, we find $e^{0.24} \approx 1.27125$. Therefore,

$$A \approx 50,000 \times 1.27125 = 63,562.50.$$

The interpretation is straightforward. Once the growth convention is continuous, the exponential formula gives the consistent future value corresponding to that convention.

Common Mistake

Failing example: mismatched rate conventions Suppose a student hears that the annual rate is “six per cent” and then assumes that any growth formula can be used interchangeably. If the student computes $50,000(1.06)^4$, the answer is approximately HK\$63,123.85 rather than the continuous-model value above. This does not mean one answer is mathematically careless in itself. It means the student has silently switched from a continuous growth convention to an effective annual compounding convention. The mistake lies in pretending that the two conventions are the same. When the formula changes, the meaning of the rate must remain consistent.

5.2 E2. Continuous discounting**Key Definition / Result**

If a future payment F due at time t is discounted at continuous annual rate r , then its present value is

$$PV = Fe^{-rt}.$$

The negative exponent signals movement backward from the future to the present.

Worked Example

Present value under continuous discounting (Core) Suppose an insurer expects to pay HK\$1,200,000 in five years and the continuous annual discount rate is 4%. The present value is

$$PV = 1,200,000e^{-0.20}.$$

Because $e^{-0.20} \approx 0.8187308$, the present value is approximately HK\$982,476.96. In plain language, if the relevant continuous discount rate is 4%, then a present fund of about HK\$982,476.96 is financially equivalent to the obligation of paying HK\$1.2 million in five years.

Investigation / Activity

The meaning of the negative exponent in liability valuation. An insurer must pay HK\$1,200,000 in five years for a catastrophe-related claim. The valuation team uses $PV = 1,200,000e^{-0.04 \times 5}$. Explain, in complete sentences, why the exponent is negative. Then explain why a continuous model can be attractive when market conditions are thought of as evolving smoothly rather than in isolated annual jumps.

AI Exploration Prompt

A suitable AI exploration prompt would be: “Act as a chief risk officer. Explain to a junior actuary why continuous discounting, written as e^{-rt} , can be a useful benchmark for valuing future insurance liabilities. Give a realistic catastrophe-liability scenario and explain the interpretation in simple language.”

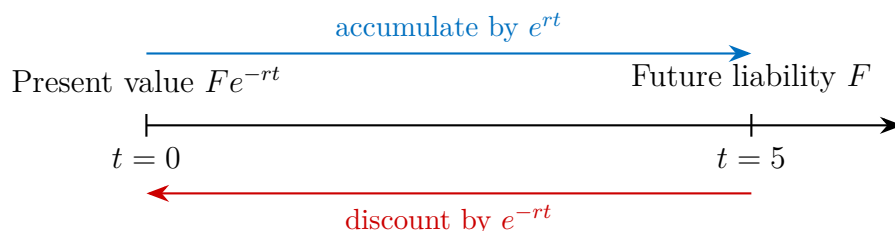


Figure 7: Continuous accumulation and continuous discounting as inverse translations across time.

5.3 E3. Equivalent effective rates and the natural logarithm

If r is a continuously compounded annual rate and i is the equivalent effective annual rate, then

$$1 + i = e^r.$$

Therefore,

$$i = e^r - 1 \quad \text{and} \quad r = \ln(1 + i).$$

The symbol $\ln(x)$ denotes the natural logarithm, meaning the logarithm to base e . It is the inverse function of the exponential expression e^x .

Table 6: A concise conversion table between effective and continuous annual rates

Given quantity	Required quantity	Formula
Continuous rate r	Effective annual rate i	$i = e^r - 1$
Effective annual rate i	Continuous rate r	$r = \ln(1 + i)$

Worked Example

Converting an effective annual rate into a continuous rate (Core) If the effective annual rate is 9%, then the one-year growth factor is 1.09. The equivalent continuous rate satisfies $e^r = 1.09$, so

$$r = \ln(1.09) \approx 0.08618.$$

Thus the equivalent continuous rate is approximately 8.618%. The numerical value is lower than 9%, but both conventions produce the same one-year accumulation factor when applied correctly.

6 F. Log Returns and the Value of Continuous-Time Reasoning

6.1 F1. Two ways to measure financial change

If a price moves from P_0 to P_1 , the ordinary percentage return is

$$\frac{P_1 - P_0}{P_0},$$

while the continuously compounded or log return is

$$\ln\left(\frac{P_1}{P_0}\right).$$

The two measures are similar when changes are small, but they are not identical. The log-return form is especially important because it adds naturally over time.

Worked Example

Ordinary return compared with log return (Core) Suppose a stock rises from 100 to 112. The ordinary return is

$$\frac{112 - 100}{100} = 0.12 = 12\%.$$

The log return is

$$\ln\left(\frac{112}{100}\right) = \ln(1.12) \approx 0.11333.$$

Thus the ordinary return is 12%, while the log return is approximately 11.33%. The values are close, but they measure growth under different conventions.

6.2 F2. Why log returns are often preferred in professional modelling

The key advantage of log returns is additivity. If a price changes by successive factors $g_1, g_2, \dots, g_n$, the overall growth factor is the product $g_1 g_2 \cdots g_n$. Taking logarithms converts this product into a sum:

$$\ln(g_1 g_2 \cdots g_n) = \ln(g_1) + \ln(g_2) + \cdots + \ln(g_n).$$

This property makes multi-period analysis cleaner and often more interpretable in modelling.

Technology Check (Google Gemini / Spreadsheet / R statistical software)

Comparing ordinary and log returns using software (Insight) Create a short sequence of prices in a spreadsheet or in R and compute both the ordinary return and the log return for each adjacent pair. Then compare the sum of the log returns with the logarithm of the overall price ratio from the first observation to the last.

```
P0 <- c(100,102,99,103)
P1 <- c(102,99,103,105)
ordinary <- (P1-P0)/P0
logr <- log(P1/P0)
data.frame(P0,P1,ordinary,logr)
```

6.3 F3. Solving for time and rate in the continuous model

Starting from

$$A = Pe^{rt},$$

one can solve for time or rate by applying the natural logarithm. This gives

$$t = \frac{\ln(A/P)}{r} \quad \text{and} \quad r = \frac{\ln(A/P)}{t}.$$

The formulas are elegant because the natural logarithm is the inverse of the exponential function.

Worked Example

How long does it take for a fund to double? (Insight) Suppose a reserve fund grows at continuous annual rate 7%. To find the doubling time, set $A = 2P$. Then

$$2 = e^{0.07t}.$$

Taking natural logarithms gives $\ln(2) = 0.07t$, so

$$t = \frac{\ln(2)}{0.07} \approx 9.90.$$

Thus the reserve fund takes approximately 9.90 years to double under the stated continuous annual rate.

7 G. Investigation, Activity, and Discussion

Investigation / Activity

Investigation 1: Does more frequent compounding matter in practice?

Choose a principal amount such as HK\$100,000 and compare annual, quarterly, monthly, daily, and continuous compounding at the same quoted annual rate. Present your results in a table and then write a paragraph discussing whether the numerical differences are economically small or large in the chosen context. Your discussion should use complete sentences and should refer to the original purpose of the investment.

Investigation / Activity

Investigation 2: Asset choice for a reserve fund. Imagine that a school, charity, or insurer expects a large payment in three years. Compare a short-term bond strategy, an equity-heavy strategy, and a mixed strategy. Write a short advisory note in complete sentences explaining the trade-off among expected return, liquidity, volatility, and suitability for the liability. The goal is not to identify the highest return in isolation, but to identify the strategy that is most appropriate for the purpose.

Investigation / Activity

Investigation 3: Why do log returns add? Build a sequence of at least four price changes. Compute the ordinary return and the log return in each period. Then compare the sum of the log returns with the logarithm of the overall price ratio from the beginning to the end of the sequence. In a short paragraph, explain why this additivity is useful in modelling and why ordinary returns do not behave in the same way.

8 Review Set A (Foundation)

Review Set A

Answer each question in full sentences where interpretation is required, and show explicit working where calculation is required.

- A1:** Explain why financial markets are important for risk managers and actuaries.
- A2:** Distinguish carefully between a primary market and a secondary market.
- A3:** Explain why liquidity matters to an insurer or another institution facing future obligations.
- A4:** Describe the difference between debt, equity, and derivatives in terms of claim structure.
- A5:** Explain, in complete sentences, why a shareholder is called a residual claimant.
- A6:** A fund of HK\$120,000 is invested at a quoted annual rate of 6%, compounded monthly, for three years. Compute the future value.
- A7:** A liability of HK\$500,000 is due in five years. Find its present value at effective annual discount rate 4%.
- A8:** Compute $\left(1 + \frac{1}{2}\right)^2$, $\left(1 + \frac{1}{12}\right)^{12}$, and $\left(1 + \frac{1}{365}\right)^{365}$, and explain the pattern you observe.
- A9:** A three-year bond has face value HK\$1,000, annual coupon rate 5%, and market yield 4%. Find its price and explain why the price is above or below face value.
- A10:** A stock is expected to pay a dividend of HK\$2.40 next year. If the required return is 8% and the expected constant growth rate is 3%, find the price using the constant-growth model and interpret the result.

- A11:** A fund of HK\$50,000 grows at continuous annual rate 6% for four years. Find the future value.
- A12:** Convert an effective annual rate of 9% into an equivalent continuously compounded annual rate.
- A13:** A stock rises from 100 to 112. Find the ordinary return and the log return, then explain why the two values differ.

9 Review Set B (More Challenging)

Review Set B

The following questions ask for stronger interpretation, synthesis, and transfer.

- B1:** A student says, “If two investments have the same stated annual rate, then they must produce the same future value.” Explain carefully why this statement can be false.
- B2:** Describe a realistic situation in which an asset may have high theoretical value and yet create serious risk because of low liquidity.
- B3:** A Treasury bill with face value HK\$10,000 is sold for HK\$9,750 and matures in 180 days. Compute both the discount yield and the bond-equivalent yield, and explain why they differ.
- B4:** A bond with face value HK\$1,000 pays annual coupons of HK\$60 for four years and repays principal at maturity. If the market yield is 5%, find the bond price and explain whether it should trade at a premium or a discount.
- B5:** A stock is expected to pay dividend $D_1 = 3.20$ next year, with required return 10% and constant growth rate 4%. Find the price under the constant-growth model, then explain how the answer changes if the required return rises while growth stays the same.
- B6:** Explain why diversification reduces the variability of an average portfolio outcome but cannot fully eliminate systemic risk.
- B7:** Explain why a large pool of exposures can still perform poorly if correlation becomes very high in stressed conditions.
- B8:** A reserve fund grows according to $A(t) = 750,000e^{0.045t}$. Find the value after eight years and the time needed to reach HK\$1,200,000.
- B9:** An insurer expects to pay HK\$1,200,000 in five years. Compute the present

value at continuous annual discount rate 4%, and interpret the result in plain language.

- B10:** A price rises by 20% in one year and then falls by 20% in the next year. Find the two one-year log returns and their sum. Then compare them with the overall ordinary return over the two years.
- B11:** Explain in complete sentences why $r = \ln(1 + i)$ is the correct conversion from an effective annual rate to a continuous annual rate.
- B12:** Write a short paragraph titled *Why future actuaries should care about Euler's number e* .
- B13:** Use technology to compare annual, monthly, daily, and continuous compounding for HK\$100,000 at quoted annual rate 8% for one year. Present a table of results and comment on whether the differences are practically large or small.

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Key Definition / Result

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